

VCS Task -Sudir

Task Description:

Establish a new directory, populate it with script files, initiate an empty repository on GitHub, convert the local directory into a Git repository, and link it to GitHub for pushing the code into the repository. Perform merge, rebase, stash commands in following github repo.

Techstacks needs to be used :

- Shell (AWS, WSL, Vbox)
- Git (Gitbash)

How do I submit my work?

- Push all your work files to GitHub (Code & O/P screenshot images must).
- Submit your URLs in the portal.

Terms and Conditions?

- You agree to not share this confidential document with anyone.
- You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.
- We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.

NOTE: Any violation of Terms and conditions is strictly prohibited. You are bound to adhere to it.

Login to your Server

```
sudir@sonarjen:~$ whoami  
sudir
```

Confirm Git Installation & Configuration:

```
15 sudo apt install git -y  
16 git --version  
17 git config --global user.name  
18 git config --global user.name "sudir"  
19 git config --global user.email "sudir@gmail.com"  
20 git config --list  
21 history
```

Create Project Directory

Action: Navigate to the directory where you want to create your project and then create a new directory.

Command:

- mkdir my_project
- cd my_project

```
sudir@sonarjen:~$ mkdir my_project  
sudir@sonarjen:~$ cd my_project  
sudir@sonarjen:~/my_project$ pwd  
/home/sudir/my_project  
sudir@sonarjen:~/my_project$
```

Create Script Files

Action: Create some basic script files (e.g., file1.sh and file2.sh).

Command:

- touch file1.sh file2.sh

```
sudir@sonarjen:~/my_project$ ls  
file1.sh  file2.sh
```

Edit Script Files

Action: Edit the files using a text editor (e.g., nano, vim).

Example:

- vim file1.sh

```
sudir@sonarjen:~/my_project$ vim file1.sh
sudir@sonarjen:~/my_project$ cat file1.sh
#!/bin/bash
echo "Hello from file1.sh"
```

Initialize Git Repository Locally

Action: Initialize a Git repository in your project directory.

Command:

- git init

```
Initialized empty Git repository in /home/sudir/my_project/.git/
sudir@sonarjen:~/my_project$ ll
total 16
drwxrwxr-x 3 sudir sudir 4096 Sep 22 11:39 ./
drwxr-x--- 8 sudir sudir 4096 Sep 22 11:38 ../
-rw-rw-r-- 1 sudir sudir  40 Sep 22 11:38 file1.sh
-rw-rw-r-- 1 sudir sudir   0 Sep 22 11:36 file2.sh
drwxrwxr-x 6 sudir sudir 4096 Sep 22 11:39 .git/
sudir@sonarjen:~/my_project$
```

Check Git Status

Action: Check the status of your repository to confirm it's initialized.

Command:

- git status

```
sudir@sonarjen:~/my_project$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.sh
        file2.sh

nothing added to commit but untracked files present (use "git add" to track)
sudir@sonarjen:~/my_project$
```

Add Files to Git

Action: Add your files to the staging area.

Command:

- `git add .`

Files are now staged

```
sudir@sonarjen:~/my_project$ git add .
sudir@sonarjen:~/my_project$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   file1.sh
        new file:   file2.sh

sudir@sonarjen:~/my_project$
```

Commit the Changes

Action: Commit the files with a meaningful message.

Command:

- `git commit -m "Initial commit with script files"`

```
sudir@sonarjen:~/my_project$ git commit -m "Initial commit with script files"
[master (root-commit) a22c546] Initial commit with script files
Committer: sudir <sudir@sonarjen>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

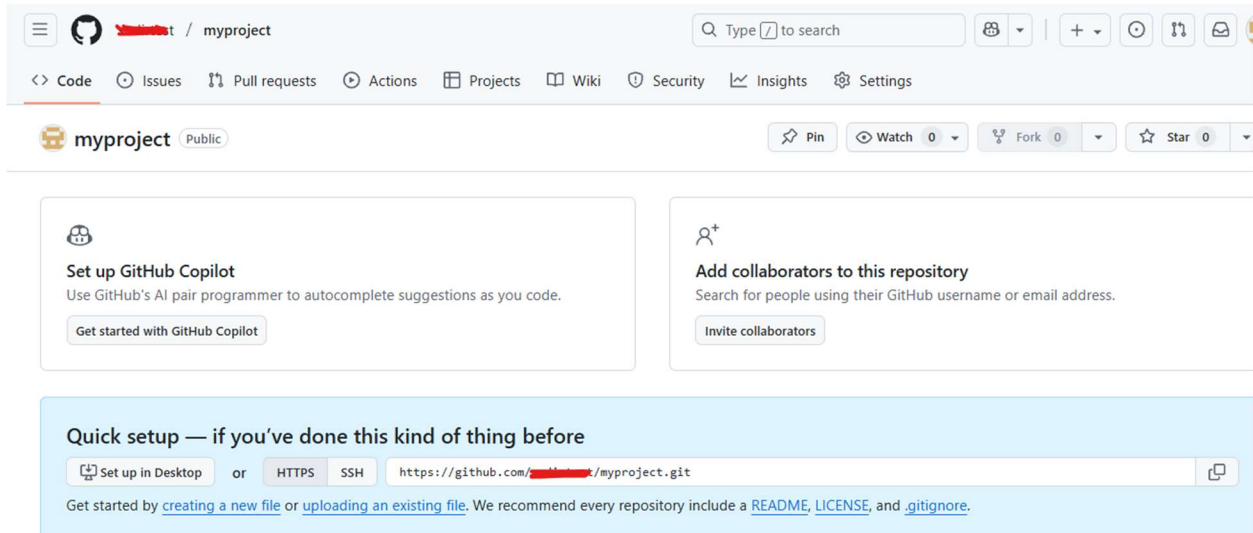
    git commit --amend --reset-author

2 files changed, 3 insertions(+)
create mode 100644 file1.sh
create mode 100644 file2.sh
```

Create a Remote GitHub Repository

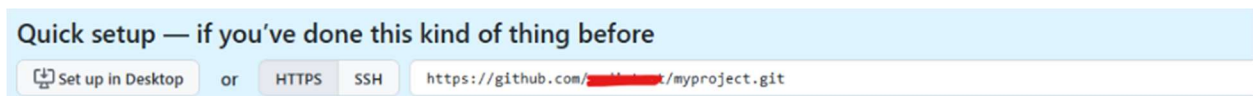
Go to GitHub

Action: Log in to GitHub and create a new repository (leave it empty, without a README or .gitignore file).



Copy the GitHub Repository URL

Action: After creating the repository, copy the repository URL (HTTPS or SSH).



Link Local Repository to GitHub

Add Remote Origin

Action: Link your local repository to GitHub using the git remote add command.

Command:

- `git remote add origin <repository_URL>`

```
sudir@sonarjen:~/my_project$ git remote add origin https://github.com/sudirstest/myproject.git
sudir@sonarjen:~/my_project$
```

Push Code to GitHub

Action: Push your local repository to GitHub.

Create a ssh key using **ssh-keygen** and copy the **public key** and store it in repo **Deploy keys**.

Command:

- `git push -u origin master`

```
sudir@sonarjen:~/my_project$ git remote -v
origin  git@github.com:sonarjen/myproject.git (fetch)
origin  git@github.com:sonarjen/myproject.git (push)
sudir@sonarjen:~/my_project$ git push -u origin master
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 293 bytes | 18.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sonarjen/myproject.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
sudir@sonarjen:~/my_project$
```

Git Operations (Merge, Rebase, Stash)

Merge Branches

Create a Feature Branch

Action: Create a new branch for the feature you're working on.

Command:

- `git checkout -b feature-branch`

```
sudir@sonarjen:~/my_project$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
sudir@sonarjen:~/my_project$
```

```
sudir@sonarjen:~/my_project$ git branch
* feature-branch
  master
sudir@sonarjen:~/my_project$
```

Make Changes and Commit

Action: Edit one of your files (e.g., file1.sh) and commit the changes.

Command:

- `git add .`
- `git commit -m "Changes in feature-branch"`
- `git push origin feature-branch`

```
sudir@sonarjen:~/my_project$ vim file1.sh
sudir@sonarjen:~/my_project$ cat file1.sh
#!/bin/bash
echo "making changes in file1.sh"

sudir@sonarjen:~/my_project$
```

```
nothing to commit, working tree clean
sudir@sonarjen:~/my_project$ vim file1.sh
sudir@sonarjen:~/my_project$ cat file1.sh
#!/bin/bash
echo "making changes in file1.sh"

sudir@sonarjen:~/my_project$ git add .
sudir@sonarjen:~/my_project$ git commit -m "Changes in feature-branch"
[feature-branch 4167e30] Changes in feature-branch
 1 file changed, 1 insertion(+), 1 deletion(-)
sudir@sonarjen:~/my_project$
```

Switch to the master Branch and Merge

Action: Switch back to the master branch and merge the feature-branch.

Command:

- `git checkout master`
- `git merge feature-branch`


```
sudir@sonarjen:~/my_project$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
sudir@sonarjen:~/my_project$ git branch
  feature-branch
* master
sudir@sonarjen:~/my_project$
```

```
sudir@sonarjen:~/my_project$ git merge feature-branch
Updating a22c546..4167e30
Fast-forward
 file1.sh | 2 +
 1 file changed, 1 insertion(+), 1 deletion(-)
sudir@sonarjen:~/my_project$
```

```
sudir@sonarjen:~/my_project$ git push origin feature-branch
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 326 bytes | 326.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-branch' on GitHub by visiting:
remote:   https://github.com/[redacted]/myproject/pull/new/feature-branch
remote:
To github.com:[redacted]:myproject.git
 * [new branch]      feature-branch -> feature-branch
sudir@sonarjen:~/my_project$
```

 **Merged**

Changes in feature-branch #1
[redacted] merged 1 commit into **master** from **feature-branch**  now

  Changes in feature-branch

  [redacted] merged commit 2408964 into **master** now

 **Pull request successfully merged and closed**
You're all set — the **feature-branch** branch can be safely deleted.

Rebase

Rebase a Feature Branch on Master

Action: Rebase the feature branch on master before pushing it.

Command:

- `git checkout feature-branch`
- `git rebase master`

```
sudoir@sonarjen:~/my_project$ git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)
sudoir@sonarjen:~/my_project$ vim file1.sh
```

```
sudoir@sonarjen:~/my_project$ git add .
sudoir@sonarjen:~/my_project$ git commit -m "rebase"
[master b5a8259] rebase
 1 file changed, 1 insertion(+), 1 deletion(-)
sudoir@sonarjen:~/my_project$ git checkout feature-branch
Switched to branch 'feature-branch'
sudoir@sonarjen:~/my_project$ git rebase master
Successfully rebased and updated refs/heads/feature-branch.
sudoir@sonarjen:~/my project$
```

Make Changes and Stash

Action: Make some changes to files and stash them temporarily.

Command:

- `git stash`

```
sudoir@sonarjen:~/my_project$ git branch
* feature-branch
  master
sudoir@sonarjen:~/my_project$ vim file1.sh
sudoir@sonarjen:~/my_project$ cat file1.sh
#!/bin/bash
echo "making changes for stash file1.sh"

sudoir@sonarjen:~/my_project$ git stash
Saved working directory and index state WIP on feature-branch: b5a8259 rebase
```

Apply Stashed Changes

Action: After switching branches, apply the stashed changes.

Command:

git stash apply

```

sudir@sonarjen:~/my_project$ echo "Temporary line for stash dem" >> file1.sh
sudir@sonarjen:~/my_project$ echo "Adding temp line" >> file2.sh
sudir@sonarjen:~/my_project$ git status
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   file1.sh
    modified:   file2.sh

no changes added to commit (use "git add" and/or "git commit -a")
sudir@sonarjen:~/my_project$ git stash push -m "Demo stash changes"
Saved working directory and index state On feature-branch: Demo stash changes
sudir@sonarjen:~/my_project$ git stash push -m "Demo stash changes"
No local changes to save
sudir@sonarjen:~/my_project$ git stash list
stash@{0}: On feature-branch: Demo stash changes
stash@{1}: On feature-branch: Demo stash changes
stash@{2}: WIP on feature-branch: b5a8259 rebase
sudir@sonarjen:~/my_project$ git stash push -m "Demo stash changes"
No local changes to save
sudir@sonarjen:~/my_project$ git stash apply stash@{0}
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   file1.sh
    modified:   file2.sh

no changes added to commit (use "git add" and/or "git commit -a")
sudir@sonarjen:~/my_project$ git add .
sudir@sonarjen:~/my_project$ git commit -m "Re-applied stashed changes"
[feature-branch 27d9e90] Re-applied stashed changes
 2 files changed, 2 insertions(+)
sudir@sonarjen:~/my_project$ git stash drop stash@{0}
Dropped stash@{0} (0b0861ee2977bce8bf074ceaa8623a0a19f37elf)
sudir@sonarjen:~/my_project$
```

Pushing and Pulling from GitHub

Push to GitHub

Action: Push the local changes to the GitHub repository.

Command:

- `git push -u origin feature-branch1`

```

sudir@sonarjen:~/my_project$ git checkout -b feature-branch1 origin/feature-branch1
branch 'feature-branch1' set up to track 'origin/feature-branch1'.
Switched to a new branch 'feature-branch1'
sudir@sonarjen:~/my_project$ git branch
  feature-branch
* feature-branch1
  master
sudir@sonarjen:~/my_project$ touch file1 file2
sudir@sonarjen:~/my_project$ ll
total 16
drwxrwxr-x 3 sudir sudir 4096 Sep 22 12:58 ./
drwxr-x--- 9 sudir sudir 4096 Sep 22 12:58 ../
-rw-rw-r-- 1 sudir sudir   0 Sep 22 12:58 file1
-rw-rw-r-- 1 sudir sudir  48 Sep 22 12:58 file1.sh
-rw-rw-r-- 1 sudir sudir   0 Sep 22 12:58 file2
-rw-rw-r-- 1 sudir sudir   0 Sep 22 12:51 file2.sh
drwxrwxr-x 7 sudir sudir 4096 Sep 22 12:58 .git/
sudir@sonarjen:~/my_project$ git add .
sudir@sonarjen:~/my_project$ git commit -m "Your commit message"
[feature-branch1 131ldf9] Your commit message
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file1
 create mode 100644 file2
sudir@sonarjen:~/my_project$ git push -u origin feature-branch1
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 272 bytes | 136.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-branch1' on GitHub by visiting:
remote:   https://github.com/[redacted]/myproject/pull/new/feature-branch1
remote:
To github.com:[redacted]:myproject.git
 * [new branch]      feature-branch1 -> feature-branch1
branch 'feature-branch1' set up to track 'origin/feature-branch1'.

```

Create a new a file in repository and pull from local

Command:

- `git pull origin feature-branch1`

feature-branch1 myproject /

Create aaa b59e30f · now History

This branch is 2 commits ahead of, 2 commits behind master. Contribute

Name	Last commit message	Last commit date
aaa	Create aaa	now
file1	Your commit message	3 minutes ago
file1.sh	Changes in feature-branch	50 minutes ago
file2	Your commit message	3 minutes ago
file2.sh	Initial commit with script files	1 hour ago

```
sudir@sonarjen:~/my_project$ ll
total 16
drwxrwxr-x 3 sudir sudir 4096 Sep 22 12:58 ./
drwxr-x--- 9 sudir sudir 4096 Sep 22 12:58 ../
-rw-rw-r-- 1 sudir sudir  0 Sep 22 12:58 file1
-rw-rw-r-- 1 sudir sudir 48 Sep 22 12:58 file1.sh
-rw-rw-r-- 1 sudir sudir  0 Sep 22 12:58 file2
-rw-rw-r-- 1 sudir sudir  0 Sep 22 12:51 file2.sh
drwxrwxr-x 7 sudir sudir 4096 Sep 22 12:59 .git/
sudir@sonarjen:~/my_project$ git pull origin feature-branch1
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 904 bytes | 452.00 KiB/s, done.
From github.com:sudirtest/myproject
* branch          feature-branch1 -> FETCH_HEAD
   1311df9..b59e30f feature-branch1 -> origin/feature-branch1
Updating 1311df9..b59e30f
Fast-forward
   aaa | 1 +
   1 file changed, 1 insertion(+)
   create mode 100644 aaa
sudir@sonarjen:~/my_project$ ll
total 20
drwxrwxr-x 3 sudir sudir 4096 Sep 22 13:03 ./
drwxr-x--- 9 sudir sudir 4096 Sep 22 12:58 ../
-rw-rw-r-- 1 sudir sudir  7 Sep 22 13:03 aaa
-rw-rw-r-- 1 sudir sudir  0 Sep 22 12:58 file1
-rw-rw-r-- 1 sudir sudir 48 Sep 22 12:58 file1.sh
-rw-rw-r-- 1 sudir sudir  0 Sep 22 12:58 file2
-rw-rw-r-- 1 sudir sudir  0 Sep 22 12:51 file2.sh
drwxrwxr-x 7 sudir sudir 4096 Sep 22 13:03 .git/
sudir@sonarjen:~/my_project$
```

